

Auteur: Thijs Van Schel
Studierichting: Informatica
Oefeningenreeks 6E nr. 1.5

$$f(x) = \frac{1}{x-1} = (x-1)^{-1}$$

$$f(x) = (x-1)^{-1} \Rightarrow f(2) = 1$$

$$f'(x) = -1(x-1)^{-2} \Rightarrow f'(2) = -1$$

$$f''(x) = 2(x-1)^{-3} \Rightarrow f''(2) = 2$$

$$f'''(x) = -6(x-1)^{-4} \Rightarrow f'''(2) = -6$$

$$f^{(4)}(x) = 24(x-1)^{-5} \Rightarrow f^{(4)}(2) = 24$$

$$f^{(5)}(x) = -120(x-1)^{-6} \Rightarrow f^{(5)}(2) = -120$$

$$f^{(6)}(x) = 720(x-1)^{-7} \Rightarrow f^{(6)}(2) = 720$$

$$f^{(7)}(x) = -5040(x-1)^{-8} \Rightarrow f^{(7)}(2) = -5040$$

Taylorformule invullen:

$$T_7(f, 2)(x) = \frac{f(2)}{0!}(x-2)^0 + \frac{f'(2)}{1!}(x-2)^1 + \frac{f''(2)}{2!}(x-2)^2 + \frac{f'''(2)}{3!}(x-2)^3 + \dots$$

$$T_7(f, 2)(x) = \frac{1}{1}(1) + \frac{-1}{1}(x-2)^1 + \frac{2}{2}(x-2)^2 + \frac{-6}{6}(x-2)^3 + \frac{24}{24}(x-2)^4 + \frac{-120}{120}(x-2)^5 + \frac{720}{720}(x-2)^6 + \frac{-5040}{5040}(x-2)^7$$

$$T_7(f, 2)(x) = 1 - (x-2) + (x-2)^2 - (x-2)^3 + (x-2)^4 - (x-2)^5 + (x-2)^6 - (x-2)^7$$

References